Motor Execution Priming Enhances Frontocentral Sensorimotor Rhythm Modulation During Motor Imagery Using a Low-Cost EEG-Based Brain–Computer Interface

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Motor imagery (MI)-based brain–computer interfaces (BCIs) decode movement intentions through modulation of sensorimotor rhythms. However, reliable MI decoding remains challenging due to substantial inter-individual variability in MI ability and the limited sensorimotor coverage of portable EEG systems. Consumer-grade devices such as the Emotiv EPOC X do not include electrodes over canonical motor cortex locations (C3/C4), potentially reducing sensitivity to MI-related neural activity. We hypothesized that motor execution (ME) performed prior to MI would prime motor representations, enhance sensorimotor rhythm modulation, and facilitate subsequent MI-related cortical engagement.

Nine healthy adults (5 females, 4 males; 23–32 years) completed an ME-primed MI protocol involving left- and right-hand motor execution followed by motor imagery. EEG signals were recorded using the Emotiv EPOC X headset. Alpha-band (8–13 Hz) event-related desynchronization (ERD) and hemispheric lateralization were evaluated at FC5 and FC6 and compared with a preliminary MI-only dataset.

ME priming increased alpha-band ERD magnitude across frontocentral electrodes during both left- and right-hand imagery. During left-hand MI, ERD increased at FC5 (−17.62% vs. −14.25%) and FC6 (−21.68% vs. −15.34%). During right-hand MI, ERD increased at FC5 (−16.91% vs. −11.33%) and FC6 (−17.00% vs. −3.84%). In addition, hemispheric asymmetry was reduced following ME priming, with lateralization indices shifting toward zero during left-hand (−0.09 vs. 0.29) and right-hand imagery (0.01 vs. 0.17). This pattern suggests more balanced bilateral recruitment of frontocentral motor-planning networks following motor execution training.

These findings indicate that motor execution priming enhances MI-related cortical engagement and alters hemispheric recruitment patterns during motor imagery. The results further suggest that ME-assisted training may partially mitigate the impact of limited sensorimotor coverage in portable EEG systems and improve the robustness of low-cost MI-based BCIs.